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Case 26-2023: A 15-Year-Old Girl with Abdominal Pain and an Ovarian Mass

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PRESENTATION OF CASE

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N Engl J Med 2023;389:750-8.

DOI: 10.1056/NEJMcpc2211422

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Dr. Margaret R. Connolly (Surgery): A 15-year-old girl was admitted to this hospital because of abdominal pain and an ovarian mass.

The patient had been in her usual state of health until 6 weeks before the current presentation, when lower abdominal pain developed along the midline. The pain was sharp and crampy, increased with movement, and decreased with lying down. During the next 24 hours, the pain shifted from the midline to the right side. The patient was taken to the emergency department of a hospital affiliated with this hospital for evaluation.

On examination, the temporal temperature was 36.7°C, the blood pressure 125/82 mm Hg, the pulse 102 beats per minute, the respiratory rate 18 breaths per minute, and the oxygen saturation 98% while the patient was breathing ambient air. The lower abdomen had mild tenderness that was present on both sides but was greater on the right side. The abdomen was soft and nondistended. Bowel sounds were normal, and there was no rebound, guarding, or tenderness at the costovertebral angle. The white-cell count was 14,270 per microliter (reference range, 4500 to 13,500) and the hemoglobin level 10.9 g per deciliter (reference range, 12.0 to 16.0). Blood levels of electrolytes, glucose, and lactate dehydrogenase were normal, as were the results of kidney-function and liver-function tests. The urinalysis was normal. A urine test for human chorionic gonadotropin was negative. Other laboratory test results are shown in Table 1. Imaging studies were performed.

Dr. Ali Pourvaziri: Transabdominal ultrasonography with Doppler analysis revealed a complex, thick-walled cyst in the right ovary. There were multiple follicular cysts, measuring up to 1.7 cm in diameter, in the left ovary. Doppler flow was normal in both ovaries. There was a small amount of free fluid in the pelvis. The appendix was not visualized.

Dr. Connolly: Acetaminophen, ketorolac, and intravenous fluids were administered. A diagnosis of ruptured ovarian cyst was considered, and the patient was discharged home with plans to undergo repeat transabdominal ultrasonography 6 weeks later.

After the patient was discharged from the emergency department, the abdominal

Table 1. Laboratory Data.

Variable	Reference Range*	6 Wk before Current Presentation, Other Hospital	On Current Presentation, This Hospital
White-cell count (per μ l)	4500–13,500	14,270	16,120
Differential count (per μ l)			
Neutrophils	1800–8000	10,130	12,280
Lymphocytes	1500–6500	2680	2450
Monocytes	100–900	1150	1190
Eosinophils	0–400	200	50
Basophils	0–200	70	50
Hemoglobin (g/dl)	12.0–16.0	10.9	10.1
Hematocrit (%)	36.0–46.0	33.3	29.9
Platelet count (per μ l)	135,000–400,000	343,000	275,000
Erythrocyte sedimentation rate (mm/hr)	0–20	—	19
C-reactive protein (mg/liter)	0.0–8.0	5.9	27.7

* Reference values are affected by many variables, including the patient population and the laboratory methods used. They may therefore not be appropriate for all patients.

pain slowly resolved. Three days after discharge, the patient resumed competitive long-distance running. Five weeks later, and 4 days before the current presentation, she participated in a running competition. The next day, abdominal pain recurred, and the patient was unable to go to school. She took ibuprofen, but it did not alleviate the pain. She was brought to the emergency department of this hospital for evaluation.

The patient reported that the current episode of abdominal pain was similar to the episode that had occurred 6 weeks earlier. She had no fever, chills, night sweats, nausea, vomiting, anorexia, dysuria, hematuria, frequent urination, or vaginal bleeding or discharge. She had noticed an increase in her waist size during the past 2 weeks and had difficulty fitting into her pants.

The patient had no notable medical history. She took no medications and had no known allergies. She had normal growth and development; menarche had not occurred. She had received all routine childhood vaccinations. The patient lived in suburban New England with her parents and sister. She was not sexually active and did not smoke tobacco, drink alcohol, or use illicit drugs. Her parents and sister were healthy; menarche had occurred in her mother and sister when they were each 12 years of age.

On examination, the temporal temperature was

36.7°C, the blood pressure 141/89 mm Hg, and the pulse 95 beats per minute. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 18.7. The right lower abdomen was firm and distended, with mild tenderness. Bowel sounds were normal, and there was no rebound, guarding, or evidence of a discrete mass or hepatosplenomegaly on palpation. Development of pubic hair and breasts was classified as Tanner stage 4 (with stages ranging from 1 to 5 and higher stages indicating more advanced pubertal development). The white-cell count was 16,120 per microliter and the hemoglobin level 10.1 g per deciliter. Other laboratory test results are shown in Table 1. Imaging studies were performed.

Dr. Pourvaziri: Transabdominal ultrasonography with Doppler analysis revealed a heterogeneous mass, measuring 16.6 cm by 15.6 cm by 9.9 cm, that was inseparable from the right ovary with the use of probe pressure maneuvers. The mass had normal vascular flow and areas of increased echogenicity. The left ovary was normal, and there was a moderate-to-large amount of free fluid in the pelvis.

Computed tomography (CT) of the abdomen and pelvis (Fig. 1), performed after the administration of intravenous contrast material, showed a heterogeneously attenuating mass, measuring

17.8 cm by 13.6 cm by 8.6 cm, that displaced the bowel and pelvic viscera. The mass had prominent amorphous, peripherally enhancing components and internal vasculature. The right ovary was not definitively identified separately. There was a large amount of ascitic fluid. Magnetic resonance imaging (MRI) of the pelvis (Fig. 1), performed before and after the administration of intravenous contrast material, also showed a heterogeneous mass with thick, nodular, peripheral enhancement that appeared to originate from the right adnexa. The endometrial lining appeared thickened, measuring up to 22 mm in thickness.

Dr. Connolly: The patient was admitted to the hospital. A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Andrea L. Zuckerman: Six weeks before the current presentation, sharp, crampy abdominal pain developed in this otherwise healthy 15-year-old girl. Evaluation at another emergency department revealed leukocytosis. Transabdominal ultrasonography revealed a complex, thick-walled cyst in the right ovary, at the site where the patient's pain was most prominent. A presumptive diagnosis of ruptured ovarian cyst was considered.

Transabdominal ultrasonography did not show evidence of an obstructed müllerian anomaly, and the presence of normal blood flow to the ovaries on Doppler analysis made adnexal torsion unlikely. Although the appendix was not visualized on ultrasonography, a diagnosis of appendicitis was unlikely in the absence of nausea, vomiting, or anorexia. The patient was not sexually active. A negative urine pregnancy test ruled out ectopic pregnancy, and findings on imaging were not consistent with a tubo-ovarian abscess. A physiologic ovarian cyst that enlarges rapidly or ruptures can cause pain, but such cysts typically have thin walls. Physiologic ovarian cysts either resolve over time or stay the same, and repeat imaging was planned on discharge from the emergency department.

The abdominal pain initially resolved but recurred 6 weeks later, and the patient presented for evaluation in the emergency department of this hospital. Imaging studies revealed the development of a large, heterogeneously attenuating mass that displaced the bowel and pelvic viscera. Given

this clinically significant new finding, I will focus my differential diagnosis on ovarian neoplasms, including epithelial-cell tumors, germ-cell tumors, and sex cord–stromal tumors (Fig. 2).

EPITHELIAL-CELL TUMORS

Epithelial ovarian tumors may be malignant, benign, or borderline.^{1,2} On ultrasonography, these tumors are characterized by masses with nodular or papillary solid components, thick septa, and ascites. Although malignant epithelial ovarian tumors are the most common ovarian cancers overall, they occur rarely in adolescents, especially in girls younger than 17 years of age.^{3,4} This patient's age makes a malignant epithelial ovarian tumor unlikely, but the diagnosis remains a possibility in this case.

The most common benign epithelial ovarian tumors are serous and mucinous cystadenomas. Mucinous cystadenomas tend to be large and multiloculated, and serous cystadenomas tend to be bilateral and thin-walled. Neither of these tumors is likely in this patient, given the imaging characteristics of the mass, the presence of ascites, and the rapid growth of the mass over a period of 6 weeks.

Borderline epithelial ovarian tumors, also called tumors of low malignant potential, are most likely to occur in patients older than 40 years of age. These tumors can be large, with a median diameter of 14 cm. Patients most commonly present with pelvic pain, pelvic pressure, and abdominal distention. Although this patient had abdominal distention in the presence of a mass measuring nearly 18 cm in diameter, her age makes a borderline ovarian tumor unlikely.

GERM-CELL TUMORS

Germ-cell tumors are the most common tumors found in female adolescents and are therefore an important consideration in this patient. These tumors arise from primitive germ cells, which have either embryonic or extraembryonic origins, and they may be benign or malignant.

The most common germ-cell tumors of the ovary are mature cystic teratomas, which are benign. Mature cystic teratomas are composed of sebaceous material, hair, cartilage, bone, or teeth.⁵ The most common malignant germ-cell tumors

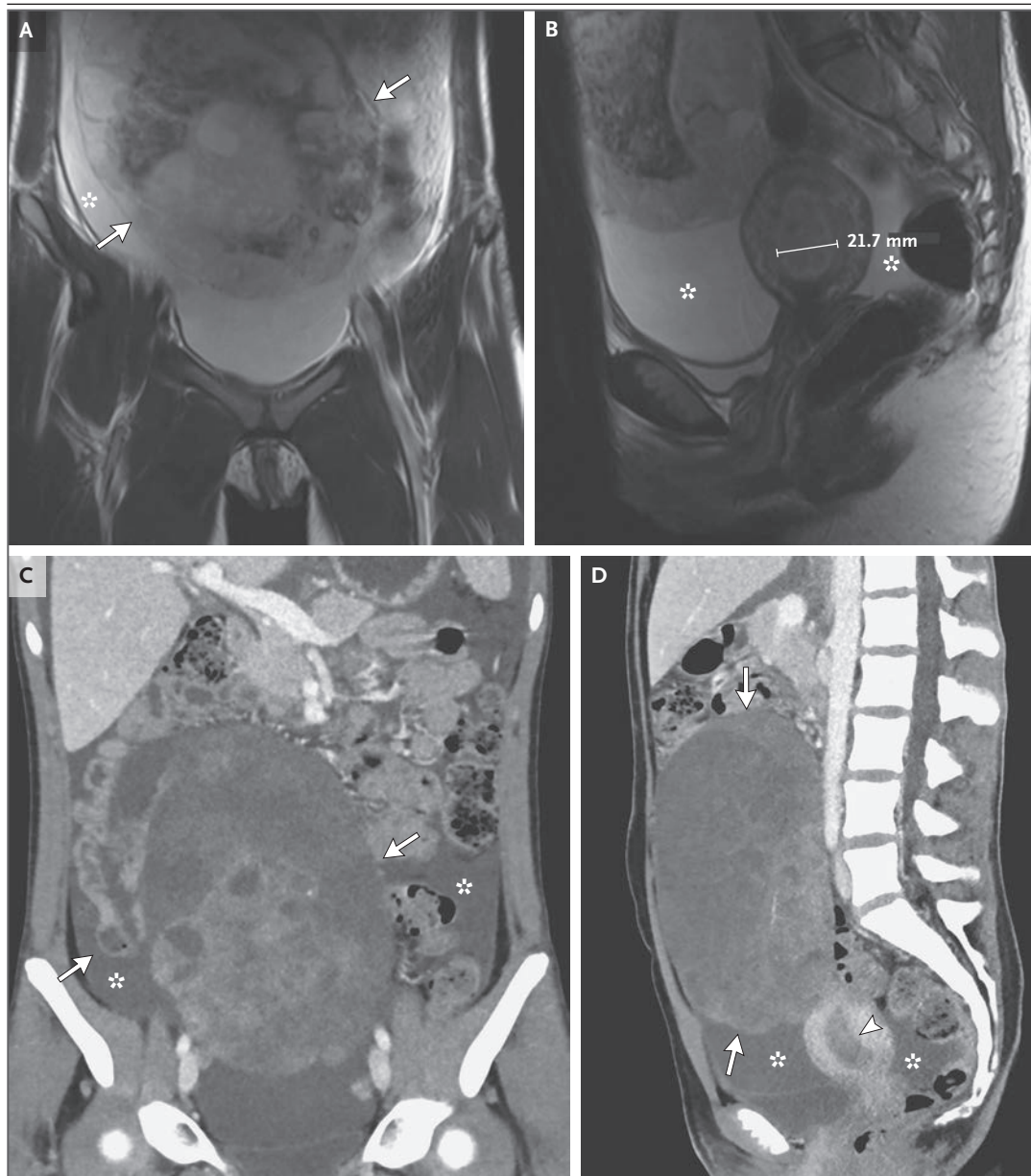


Figure 1. Imaging Studies.

MRI of the pelvis was performed. A coronal T2-weighted image (Panel A) shows a large, heterogeneous, round lesion (arrows). The lesion has internal solid, mixed hypointense and hyperintense signal intensity and multiple internal cystic foci. A sagittal T2-weighted image (Panel B) shows endometrial hypertrophy (22 mm in thickness). The lesion is only partially visible anterosuperior to the uterus and the large-volume ascites (Panels A and B, asterisks). CT of the abdomen and pelvis was performed after the administration of intravenous contrast material. Coronal and sagittal reconstruction images (Panels C and D, respectively) show the full extent of the large heterogeneous lesion (arrows). The lesion has areas of nodular, solid enhancement. Large-volume ascites in the abdomen and pelvis is also noted (Panels C and D, asterisks), along with endometrial hypertrophy (Panel D, arrowhead). There is no evidence of peritoneal thickening or deposits.

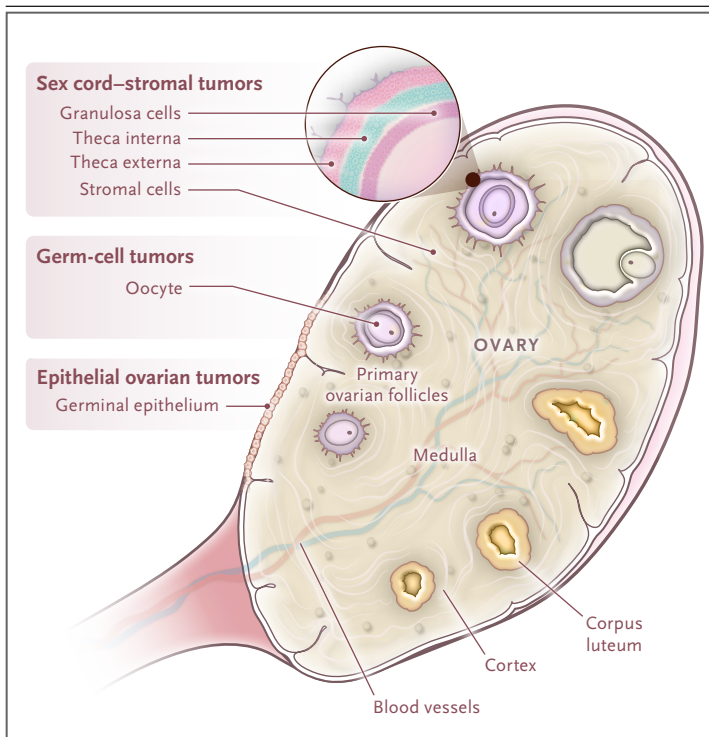


Figure 2. Origins of Ovarian Tumors.

Epithelial cells are superficial cells overlying the ovary that aid in ovulation; ovarian cancer is most likely to originate from epithelial cells. Germ cells are located within the ovary, develop into oocytes, and produce hormones. Although the most common germ-cell tumor is mature cystic teratoma, which is benign, essentially all other germ-cell tumors have malignant potential. Stromal cells make up the tissue that holds the ovary together and activates primordial follicles; most tumors that originate from sex cord or stromal cells, such as granulosa-cell tumors, produce hormones.

of the ovary are immature teratomas, which account for 35% of all malignant germ-cell tumors. Immature teratomas contain immature tissue, most often immature neural tissue, which differentiates them from mature cystic teratomas.⁶ The mass in this patient did not contain fat or calcifications, which would be expected with both of these tumor types.

The second most common malignant germ-cell tumors of the ovary are dysgerminomas, which occur bilaterally in approximately 15% of cases.⁷ These tumors are composed of immature germ cells, and approximately one third behave aggressively. Dysgerminomas can produce estrogen or testosterone, and patients may have an elevated blood level of human chorionic gonadotropin. Patients almost always have an elevated blood level of the liver enzyme lactate dehydrogenase.⁷

Although this patient's blood lactate dehydrogenase level obtained at this hospital has not been described, the mass did not have the regions of necrosis, hemorrhage, or speckled calcifications that would be expected with dysgerminomas.

The third most common malignant germ-cell tumors of the ovary are the highly aggressive endodermal sinus tumors, which are also known as yolk-sac tumors because the histologic features resemble yolk sacs. These tumors have rapid growth with extensive intraperitoneal dissemination. Patients typically have an elevated blood level of alpha-fetoprotein, which is normally made by the fetal liver and yolk sac; the blood lactate dehydrogenase level is also often elevated. Although elevated blood levels of alpha-fetoprotein and lactate dehydrogenase have not been described in this case, a yolk-sac tumor remains a possibility. Other malignant germ-cell tumors, such as embryonal carcinoma, primary choriocarcinoma, and polyembryonal carcinoma, are also possible but very rare.

SEX CORD-STROMAL TUMORS

Juvenile granulosa-cell tumors account for only 2 to 5% of all ovarian tumors, but they are the most common malignant sex cord-stromal tumors of the ovary in adolescents, occurring at a mean age of 13 years.⁸ Granulosa-cell tumors are typically unilateral and appear as septated, cystic or solid masses on imaging.⁹ They are frequently associated with ascites and intratumor hemorrhage. Patients typically present with abdominal pain. Other presenting symptoms are often related to hormones secreted by the tumor cells.

Inhibin A and B are glycoproteins secreted by granulosa cells to regulate follicle-stimulating hormone, which induces both the synthesis and the secretion of inhibin. Inhibin B is secreted by the granulosa cells of developing follicles, with the level peaking in the early-to-mid-follicular phase, whereas inhibin A is secreted by the corpus luteum, with the level peaking in the mid-luteal phase. In patients with granulosa-cell tumors that secrete inhibin B, high levels of inhibin B can suppress the normal follicle-stimulating hormone response, which in turn can lead to irregular menses or even amenorrhea.¹⁰ In addition, most patients with granulosa-cell tumors have an elevated estradiol level due to both secretion by granulosa cells and peripheral conversion of an-

drostenedione secreted by granulosa cells. Elevated estradiol levels can lead to precocious (early) puberty in children and can cause endometrial hypertrophy in postpubertal or peripubertal patients.¹¹

The most likely diagnosis in this patient — who had a large, rapidly growing tumor with associated ascites — is juvenile granulosa-cell tumor. I suspect that the granulosa-cell tumor caused an elevated inhibin B level, which led to amenorrhea, as well as an elevated estradiol level, which led to the endometrial hypertrophy identified on MRI. To establish this diagnosis, the next step would be surgical resection of the ovary with or without the fallopian tube. This may be done by means of laparoscopy or laparotomy.^{10,12} The surgical procedure includes surgical staging, as well.^{12,13}

DR. ANDREA L. ZUCKERMAN'S
DIAGNOSIS

Juvenile granulosa-cell tumor.

CLINICAL IMPRESSION AND INITIAL
MANAGEMENT

Dr. David H. Ebb: In this 15-year-old girl with a large ovarian mass and associated ascites, we were most concerned about a diagnosis of cancer. CT of the chest, performed after the administration of intravenous contrast material, revealed pleural effusions and an enhancing nodule on the pleural surface of the right hemidiaphragm. Although patients with benign ovarian neoplasms can present with effusions in both the peritoneal and the pleural cavities (Meigs' syndrome),¹⁴ the rapid growth of this patient's ovarian mass and the identification of the pleural nodule further increased our concern about cancer.

On hospital day 1, laboratory testing revealed normal blood levels of alpha-fetoprotein and lactate dehydrogenase, as well as a normal level of human chorionic gonadotropin in urine. The blood CA-125 level was 109 U per milliliter (reference value, <35) and the blood estradiol level 2708 pg per milliliter (reference range, 27 to 156). Blood levels of inhibin A and B were not immediately available.¹⁵ The normal levels of alpha-fetoprotein and human chorionic gonadotropin made a germ-cell tumor unlikely. The markedly elevated level

of estradiol, together with the presence of endometrial hypertrophy and primary amenorrhea, suggested the possibility of a functional tumor.

Regardless of the pathological diagnosis of the large ovarian mass, we thought that this patient's fulminant presentation warranted an expedited surgical resection of the massively enlarged right ovary and associated fallopian tube, along with sampling of the peritoneal fluid to look for evidence of a malignant effusion. This initial surgical effort was undertaken with the understanding that a second procedure would be necessary to sample the pleural nodule on the right hemidiaphragm for staging purposes.

Diagnostic laparotomy and right salpingo-oophorectomy were performed. Multiple peritoneal implants discovered at the time of the laparoscopic resection were sampled, along with 2 liters of bloody peritoneal fluid, and the samples were sent for cytologic analysis. At the time of intraoperative tumor mobilization, great care was taken to maintain the integrity of the capsule wall. However, the ovarian tumor was found to have clotted blood along the posterior portion of the capsule, overlying an area of capsular disruption. Although the distressed capsule could have broken intraoperatively, we suspect that the rupture may have occurred during the running competition that took place immediately before this patient's hospitalization, when the abdominal discomfort recurred.

PATHOLOGICAL DISCUSSION

Dr. Baris Boyraz: The right ovarian mass measured 18.0 cm in greatest dimension, and the outer surface was purple-gray to red. Sectioning of the mass revealed a tan-white, multilobated cut surface and a cystic cavity filled with blood (Fig. 3A). On microscopic examination, the tumor had a multinodular architecture, and the nodules contained tumor cells with predominantly sheetlike growth (Fig. 3B). Variably sized and shaped follicles with basophilic or eosinophilic secretion were focally identified (Fig. 3C). The stroma showed many areas of myxoid change. The tumor cells were polygonal and had abundant, pale, eosinophilic, variably vacuolated cytoplasm and round, hyperchromatic, primitive-appearing nuclei. The tumor cells had brisk mitotic activity, and apoptotic bodies were easily identified (Fig. 3D). These fea-

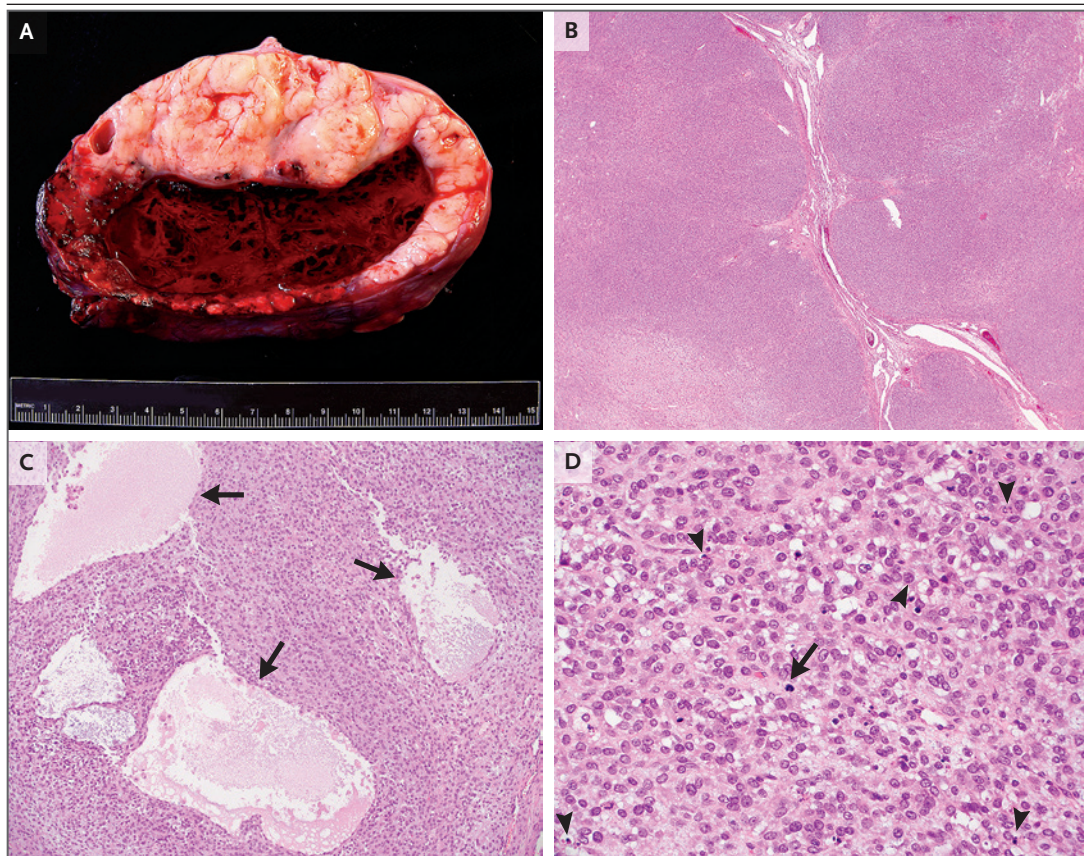


Figure 3. Ovarian Tumor Specimen.

The resected mass shows a tan-white, multilobulated cut surface and a large cystic cavity with hemorrhage (Panel A). On hematoxylin and eosin staining, the tumor has a multinodular architecture, and there is diffuse growth of tumor cells within the nodules (Panel B). Variably sized and shaped follicles with secretions are also present (Panel C, arrows). The tumor cells have abundant, eosinophilic, variably vacuolated cytoplasm and primitive-appearing nuclei, as well as abundant mitoses (Panel D, arrow) and apoptosis (Panel D, arrowheads).

tures are characteristic of juvenile granulosa-cell tumors.⁸

The designation of juvenile granulosa-cell tumors was developed approximately 50 years ago to differentiate these tumors, which are encountered in young patients, from the more familiar neoplasms that typically occur in the late reproductive and early postmenopausal years and are now referred to as adult granulosa-cell tumors. Juvenile granulosa-cell tumors are characterized by more primitive nuclei with brisker mitotic activity, a more haphazard pattern of follicles, and more abundant, typically eosinophilic cytoplasm of the neoplastic cells. The designation “juvenile” encapsulates the histologic features that typically occur in young patients, and the designation “adult” encapsulates those that typically occur in older patients, but as with so many aspects of

neoplasms, there are exceptions to these general rules. For example, adult granulosa-cell tumors can occur in children,¹⁶ and in a large study of juvenile granulosa-cell tumors, approximately 3% of the tumors occurred in patients older than 30 years of age.⁸

In this patient, analysis of the peritoneal fluid and peritoneal implants revealed inflammatory changes, without evidence of cancer.

PATHOLOGICAL DIAGNOSIS

Juvenile granulosa-cell tumor.

DISCUSSION OF MANAGEMENT

Dr. Ebb: One week after the patient's admission to this hospital, the blood inhibin A level was re-

ported as 510 pg per milliliter (reference value, <97.5) and the blood inhibin B level 58,889 pg per milliliter (reference value, <224). As noted earlier, markedly elevated levels of inhibin are characteristic features observed in patients with juvenile granulosa-cell tumors. Whereas this patient's advanced pubertal stage and endometrial hypertrophy are consistent with a high level of tumor-derived estradiol, her amenorrhea is most likely explained by elevated inhibin levels secreted by her tumor.¹⁷

One week after the patient underwent right salpingo-oophorectomy, an excisional biopsy of the pleural nodule with sampling of the pleural fluid was performed to complete her staging evaluation. Fortunately, both the pleural nodule and the pleural fluid contained no evidence of tumor dissemination, showing only the inflammatory myofibroblastic cells that had been noted in the peritoneum. Given the reassuring absence of cytologic and imaging-based evidence of tumor dissemination to the lung, liver, or serosal surfaces, we turned our attention to the management implications of the capsular rupture.

There is consensus that localized juvenile granulosa-cell tumors can be managed with surgery alone, with an excellent prognosis.¹⁸ It is equally clear that treatment with chemotherapy is indicated when these tumors have spread beyond the primary lesion to mesenteric lymph nodes, the peritoneal surface, or the liver or lung.¹⁹ Despite the use of chemotherapy, disseminated disease is associated with a much poorer long-term outcome. However, in cases in which the tumor capsule has been disrupted but there is no clear evidence of tumor spread, the use of chemotherapy as adjuvant treatment remains controversial.

After consulting with colleagues who specialize in adult gynecologic oncology, we proposed a plan

for postoperative observation with serial laboratory testing for blood levels of CA-125, estradiol, and inhibin A and B, as well as frequent surveillance with abdominal MRI and chest radiography. This plan was presented to the patient and her family in the context of an informed discussion regarding the risk of occult tumor dissemination associated with capsular rupture. After reviewing the risks and benefits of observation, we all agreed that adjuvant treatment would be deferred and introduced only after identification of laboratory or imaging-based evidence of recurrent disease.

One month after the tumor resection, the patient had resumed her athletic pursuits, and menarche occurred. Genetic profiling of her tumor revealed a somatic mutation in *GNAS* (the gene encoding the stimulatory G-protein alpha subunit), a finding that is frequently associated with juvenile granulosa-cell tumors.

During the 6 months since the tumor resection, the patient has had monthly laboratory testing, which has shown normal postoperative levels of CA-125, estradiol, and inhibin A and B. Restaging studies with abdominal MRI and chest radiography performed at 3-month intervals have revealed no evidence of recurrent disease. On the basis of our concern about the risk of relapse associated with capsular disruption, we will continue frequent laboratory testing and imaging for the next 5 years.

FINAL DIAGNOSIS

Juvenile granulosa-cell tumor.

This case was presented at Pediatric Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

We thank Dr. Jaclyn Watkins for her assistance with the pathology portion of the manuscript.

REFERENCES

- Meinhold-Heerlein I, Fotopoulou C, Harter P, et al. The new WHO classification of ovarian, fallopian tube, and primary peritoneal cancer and its clinical implications. *Arch Gynecol Obstet* 2016;293:695-700.
- Roth LM. Recent advances in the pathology and classification of ovarian sex cord-stromal tumors. *Int J Gynecol Pathol* 2006;25:199-215.
- Childress KJ, Patil NM, Muscal JA, Dietrich JE, Venkatramani R. Borderline ovarian tumor in the pediatric and adolescent population: a case series and literature review. *J Pediatr Adolesc Gynecol* 2018;31:48-54.
- Virgona C, Alaggio R, Dall'Igna P, et al. Epithelial tumors of the ovary in children and teenagers: a prospective study from the Italian TREP project. *J Pediatr Adolesc Gynecol* 2015;28:441-6.
- Zhang M, Jiang W, Li G, Xu C. Ovarian masses in children and adolescents — an analysis of 521 clinical cases. *J Pediatr Adolesc Gynecol* 2014;27(3):e73-e77.
- Tewari K, Cappuccini F, Disaia PJ, Berman ML, Manetta A, Kohler MF. Malignant germ cell tumors of the ovary. *Obstet Gynecol* 2000;95:128-33.
- Song S. Dysgerminoma. *Pathology Outlines*, September 9, 2021 (<https://www.pathologyoutlines.com/topic/ovarytumordysgerminoma.html>).
- Young RH, Dickersin GR, Scully RE. Juvenile granulosa cell tumor of the ovary: a clinicopathological analysis of 125 cases. *Am J Surg Pathol* 1984;8:575-96.
- Wang Q, Yu D, Wang F. Clinical and computed tomographic features of ovarian lesions in infants, children, and ado-

- lescents: a series of 222 cases. *J Pediatr Adolesc Gynecol* 2021;34:387-93.
10. Ye Y, Lv C, Xu S, et al. Juvenile granulosa cell tumors of the ovary. *Am J Clin Pathol* 2020;154:635-44.
 11. Wu H, Pangas SA, Eldin KW, et al. Juvenile granulosa cell tumor of the ovary: a clinicopathologic study. *J Pediatr Adolesc Gynecol* 2017;30:138-43.
 12. Guo L, Liu X-C, Chen X-Y, Xirong X, Qu Y-Q, Li B. Clinicopathological evaluation of ovarian juvenile granulosa cell tumor: is fertility-sparing surgery safe? *Reprod Dev Med* 2019;3:24-9.
 13. Al Jama FE, Al Ghamdi AA, Gasim T, Al Dakheel SA, Rahman J, Rahman MS. Ovarian tumors in children and adolescents — a clinical study of 52 patients in a university hospital. *J Pediatr Adolesc Gynecol* 2011;24:25-8.
 14. Fremed MA, Levin TL, Sun KY, Weiser DA. Ovary preservation in the treatment of childhood Meigs syndrome. *Pediatr Blood Cancer* 2015;62:2011-4.
 15. Lawrence AE, Fallat ME, Hewitt G, et al. Understanding the value of tumor markers in pediatric ovarian neoplasms. *J Pediatr Surg* 2020;55:122-5.
 16. Zaloudek C, Norris HJ. Granulosa tumors of the ovary in children: a clinical and pathologic study of 32 cases. *Am J Surg Pathol* 1982;6:503-12.
 17. Khosla D, Dimri K, Pandey AK, Mahajan R, Trehan R. Ovarian granulosa cell tumor: clinical features, treatment, outcome, and prognostic factors. *N Am J Med Sci* 2014;6:133-8.
 18. Ray-Coquard I, Morice P, Lorusso D, et al. Non-epithelial ovarian cancer: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. *Ann Oncol* 2018; 29:Suppl 4:iv1-iv18.
 19. Meisel JL, Hyman DM, Jotwani A, et al. The role of systemic chemotherapy in the management of granulosa cell tumors. *Gynecol Oncol* 2015;136:505-11.

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